

RESEARCH ARTICLE

Field Emission Control via Work Function Modulation in Semimetallic Graphene Edge Cathodes

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ABSTRACT

Graphene, the bandgapless 2D crystal, uniquely combines metal-like conductivity with an electrostatically tunable Fermi level—an ability that conventional metals lack. Here, we exploit this semimetallic property to realize a vacuum edge emitter whose current can be directly modulated through work function control rather than relying on vacuum-channel field reshaping. We demonstrate a graphene edge-emitter nanoscale vacuum transistor with an off-channel gate, enabling spatial separation of emission and control fields and achieving current saturation and low-voltage modulation across a 500 nm vacuum gap. This architecture minimizes interception and leakage while offering a structurally simple and scalable means of accessing direct Fermi-level modulation at the emission site. The device shows stable Fowler–Nordheim tunneling from 10 to 300 K, with gate-tunable emission onset and saturation current. Using small-signal parameters extracted entirely from measured I – V data, we further illustrate, through numerical analysis, that the demonstrated device characteristics are consistent with amplification behavior when placed in canonical circuit topologies. More broadly, the direct work-function modulation strategy provides a general pathway for precise control of vacuum electron emission, with potential relevance for RF, cryogenic, and radiation-resilient electronic systems.

1 | Introduction

Graphene edges have long been recognized as efficient electron emitters [1–5], benefiting from extreme field enhancement at atomically sharp edges [6, 7], exceptional carrier mobility [8], and outstanding mechanical and chemical stability. Yet, the truly distinctive feature of graphene as an emitter lies in its electrostatically tunable Fermi level [9–11], which enables dynamic work function modulation—a capability absent in conventional metallic emitters [1, 10, 11]. This dynamic tunability is particularly advantageous for nanoscale vacuum devices, where the limited footprint makes conventional gate placement problematic. For example, positioning a gate near a graphene edge can enhance field emission but inevitably intercepts a fraction of the emitted electrons. In principle, a transistor should modulate current flow without obstructing it—yet in nanoscale vacuum transistors

(NVTs), such interference is difficult to avoid. Direct work function modulation offers an alternative pathway: off-channel current control without intersecting the emission trajectory. The demand for high-speed, energy-efficient electronics has revived interest in NVTs as alternatives to solid-state devices [12–16]. By leveraging ballistic electron transport in a vacuum channel, NVTs promise ultrahigh-frequency operation and immunity to phonon scattering [13, 14, 17–19]. However, the conventional gate-in-channel geometry complicates fabrication, increases leakage, and raises operating voltages as device dimensions shrink [17].

Here, we present a graphene edge-emitter NVT featuring an off-channel direct work function control gate (DWFCG) that tunes the emitter's work function without intersecting the electron path. This spatially separated architecture suppresses leakage, simplifies fabrication, and enhances local field concentration via

reduced drain–emitter spacing [1, 5, 14, 18–20], enabling strong gate control through work function modulation rather than direct field interference.

Experimentally, we demonstrate that the DWFCG architecture provides gate-tunable current saturation—a rare yet essential trait for analog and RF operation—while maintaining stable Fowler–Nordheim (FN) tunneling from 10 to 300 K. The saturation current and emission onset voltage can be precisely set by work function tuning. Furthermore, SPICE-modeled amplifier simulations, calibrated to experimental device data, confirm voltage gain and validate the practicality of this mechanism for circuit-level applications. These results establish a scalable platform for graphene-based vacuum electronics capable of surpassing conventional solid-state devices in RF, cryogenic, and extreme-environment applications.

To clarify the technological relevance of the proposed architecture, we note that off-channel work-function tuning enables strong gate coupling without intercepting the electron stream, preserving ballistic field emission while suppressing leakage. The measured gate-dependent saturation and high output resistance provide stable biasing conditions advantageous for analog and RF electronics, allowing predictable small-signal parameter extraction. Robust operation under variant temperature conditions further suggests applicability in cryogenic or radiation-tolerant environments where semiconductor channels suffer from freeze-out or displacement damage. The simple top-gate-aligned geometry reduces parasitics compared to in-channel-gate vacuum devices and supports backend scalability for future high-frequency applications.

2 | Device Design, Fabrication, and Characterization

2.1 | Device Structure and Operation

The graphene-based NVT comprises three principal components: a monolayer graphene edge emitter integrated with the source electrode, a gate electrode, and a drain, separated by a nanometre-scale vacuum gap (Figure 1a). The atomically sharp edge of the graphene, oriented parallel to the substrate, serves as an efficient emission site, facilitating electron emission via Fowler–Nordheim tunneling under applied electric fields.

As depicted in Figure 1b, the device operates based on the principles of FN tunneling. Electrons near the Fermi level of graphene (ϵ_F) tunnel through a triangular potential barrier, the height of which is determined by the effective work function (ϕ_{gr}), defined as the energy difference between ϵ_F and the vacuum level (ϵ_{vac}). The application of an electric field bends the vacuum potential downward near the emitter surface, narrowing the tunneling barrier and exponentially increasing the emission probability, thereby enhancing the emission current.

Crucially, a DWFCG is positioned within the electrostatic screening length of the graphene edge. Rather than modulating the tunneling barrier width directly, the DWFCG tunes the local ϕ_{gr} via electrostatic gating, which alters the barrier height experi-

enced by tunneling electrons. According to our previous study, in the present device structure, the gate-induced electric field has minimal influence on the vacuum potential distribution near the edge, indicating that the emission modulation primarily originates from work function tuning rather than gate-driven changes in the channel potential [1]. This enables dynamic, gate-controlled modulation of the emission current, allowing the device to exhibit transistor-like switching behavior, a key requirement for signal amplification and logic operations in vacuum electronics.

2.2 | Device Characterization

Figure 1c shows an optical image of the fabricated DWFCG NVT with a bidirectional layout, where two emitter-gate pairs share a common drain. For DC characterization, only one side finger was selectively biased and measured. Each finger has independent source and gate terminals, avoiding asymmetry from variations in vacuum channel spacing.

The devices were fabricated on monolayer graphene transferred onto a 150 nm low-pressure chemical vapor deposition (LPCVD) Si_3N_4 dielectric layer on a silicon substrate. A scanning electron microscope (SEM) image (Figure 1d) reveals a symmetric source-gate-drain layout. Cross-sectional transmission electron microscopy (TEM) images (Figure 1e,f) confirm the vacuum channel length (~ 500 nm) and the presence of a 40 nm Al_2O_3 gate dielectric layer. Furthermore, electron energy loss spectroscopy (EELS) elemental mapping (Figure 1g) verifies the placement of the graphene layer at the emitter interface. The graphene edge is located within the electrostatic screening length of the gate, enabling effective work function modulation and gate-controlled emission [1].

To verify the structural integrity of the graphene during fabrication, Raman spectroscopy was conducted before and after dielectric deposition. As shown in Figure 1h, the spectra exhibit characteristic monolayer graphene features, including prominent G (~ 1580 cm^{-1}) and 2D (~ 2700 cm^{-1}) peaks with negligible D-band activation. This indicates minimal defect formation during atomic layer deposition (ALD). The intense and narrow 2D peak, relative to the G peak, confirms the monolayer thickness of the graphene and indicates low strain and doping levels, consistent with high-quality CVD-grown graphene [21–23].

3 | Gate-Tunable Emission Characteristics and Amplifier Operation

3.1 | Emission Current Modulation

The device was characterized in a vacuum probe station at $\sim 10^{-5}$ mbar, sufficient to suppress surface adsorption and gas interactions for reliable field emission measurements. While ultra-high vacuum may enhance long-term stability, the device maintained stable emission for at least 13 h at room temperature without degradation, demonstrating the robustness and durability of the graphene emitter for extended or high-power operation.

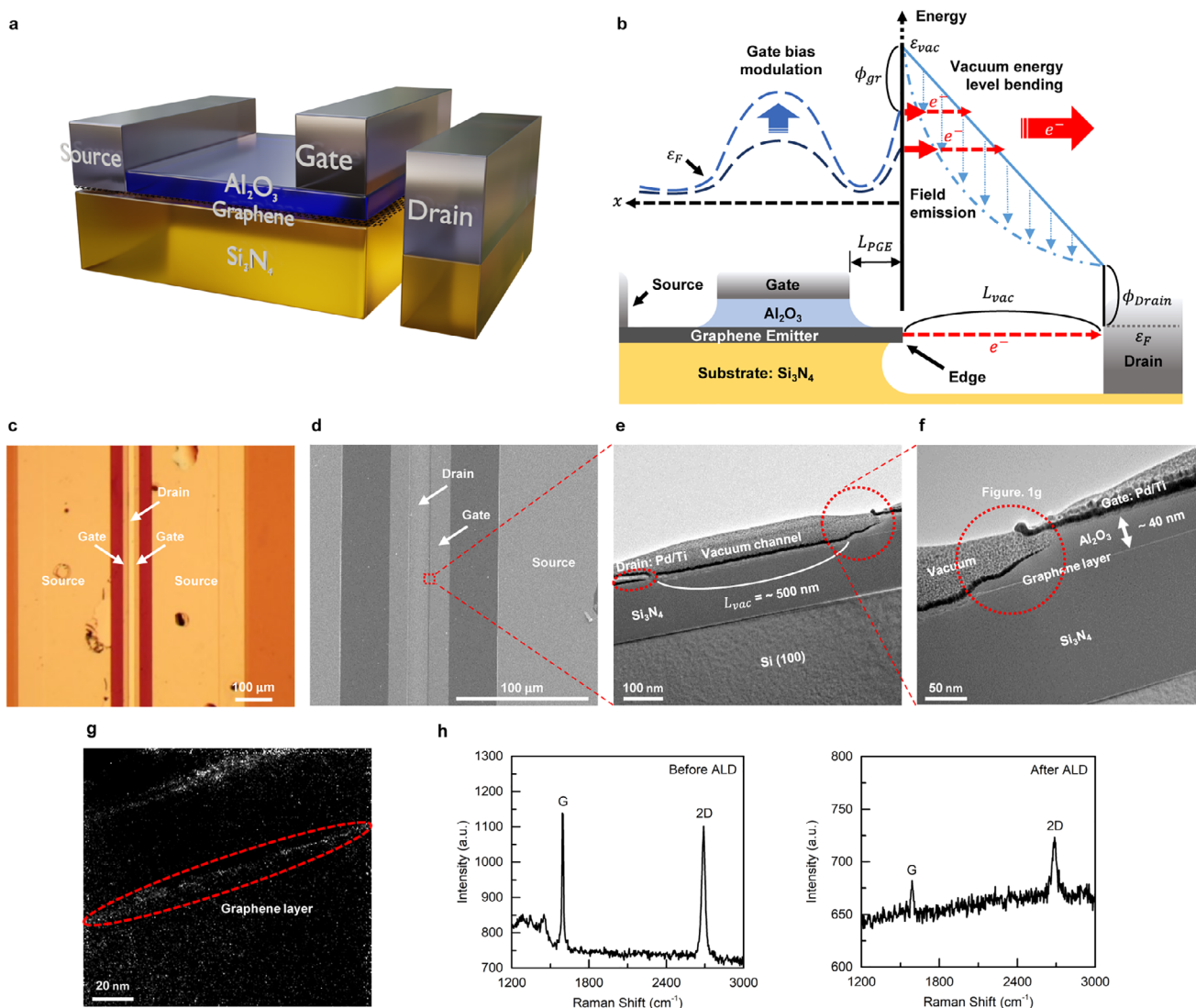


FIGURE 1 | (a) Schematic of the NVT, composed of a monolayer graphene edge emitter, a gate electrode, and a drain electrode separated by a vacuum channel. The graphene edge, oriented parallel to the substrate, serves as an electron emission source, enabling field emission via the FN tunneling mechanism. (b) Conceptual diagram of NVT operation, illustrating field emission from the graphene edge and ballistic electron transport through the vacuum channel. The gate electrode modulates the local electric field at the graphene edge, allowing for precise control of the emission current. (c) OM image of the fabricated NVT. The device exhibits a symmetric bidirectional structure where two transistors share a single drain electrode. (d) SEM image of the fabricated NVT, showing the three-terminal configuration with source, gate, and drain electrodes. Electrical measurements were conducted on only one side of the device. (e,f) TEM images of the vacuum channel and electron emission region. The vacuum channel has a length of ~ 500 nm, and the gate dielectric consists of ~ 40 nm Al_2O_3 . The device was fabricated on a 150 nm-thick Si_3N_4 layer deposited on a (100)-oriented Si wafer. (g) EELS mapping of the graphene emitter, highlighting carbon distribution and confirming the presence of graphene at the interface between the substrate and the gate dielectric. (h) Raman spectra of monolayer graphene before and after ALD of the gate dielectric. A relative decrease in the 2D/G peak ratio after ALD indicates modifications to the graphene's electronic structure, likely due to doping effects or tensile stress introduced during the ALD process. Notably, the absence of a pronounced D peak (~ 1350 cm^{-1}) suggests that the structural integrity of the monolayer graphene remains intact despite changes. All images and spectra in this figure were obtained from a single device or sample ($n = 1$). No statistical testing or p values apply, as these measurements represent deterministic physical characterization rather than population-based sampling.

Figure 2a presents the measured I – V characteristics, demonstrating clear modulation of the emission current as a function of the DWFCG bias. The gate voltage was swept from -1 V to 3 V in 1 V increments, yielding systematic changes in the emission current. This verifies the functional role of the DWFCG in modulating the effective work function of the graphene emitter and controlling the tunneling barrier. The unconventional I – V characteristics arise from the vacuum-transport nature of the device, where the

drain current is governed by Fowler–Nordheim tunneling and progressively reduced by space-charge effects at higher currents.

Two distinct regimes, characteristic of conventional vacuum diode operation, are observed in the I – V curves: a space-charge-limited (SCL) regime and a temperature-dependent (TD) regime [24, 25]. Here, “SCL” denotes space-charge-limited emission in the vacuum gap, following the Child–Langmuir framework and

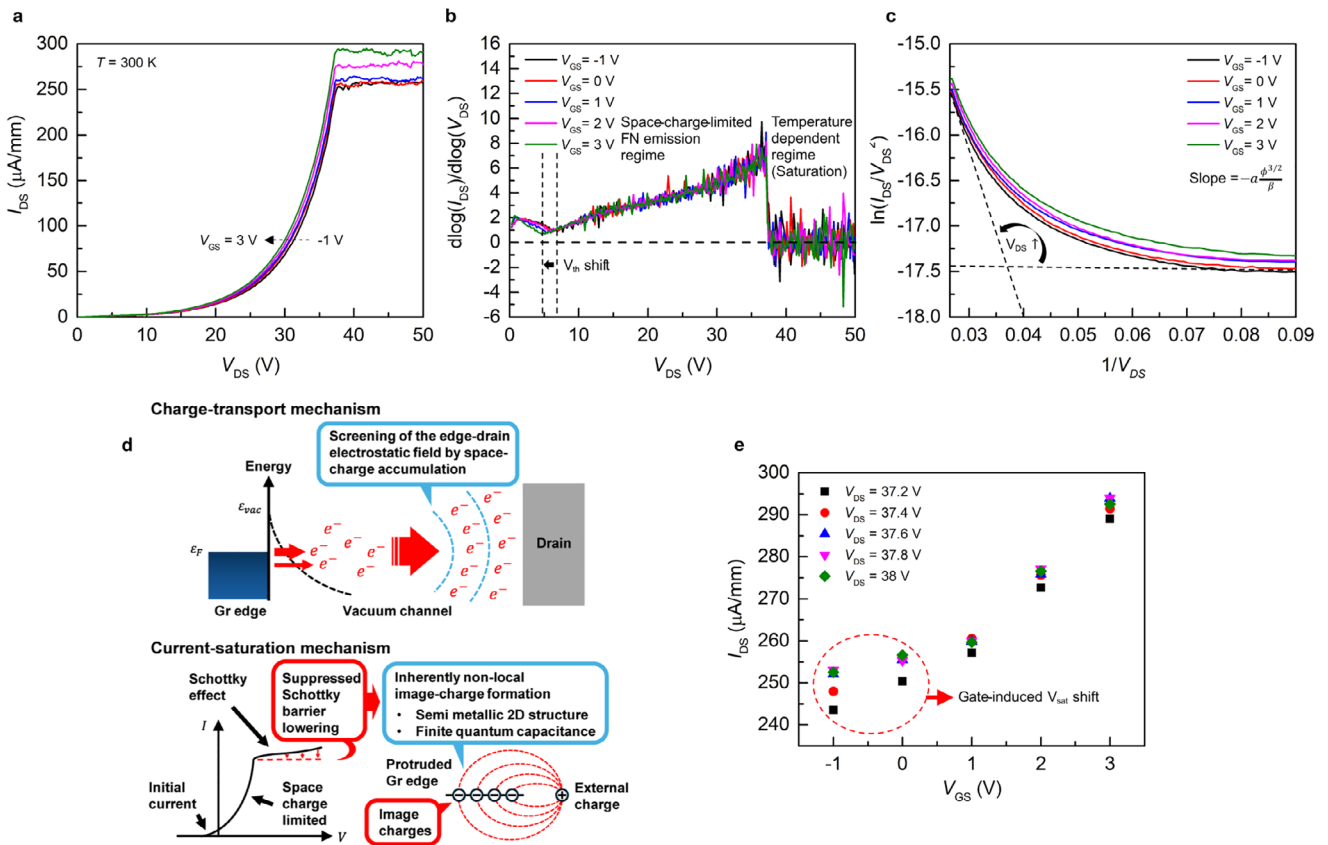


FIGURE 2 | (a) Measured I - V characteristics of the NVT, showing gate-controlled emission current modulation. The measurements were conducted at a vacuum level of 10^{-5} mbar and room temperature, with V_{GS} varied from -1 to 3 V in 1 V increments. (b) Logarithmic derivative $d\log(I_{DS})/d\log(V_{DS})$ plotted against V_{DS} , indicating distinct transport regimes: space-charge-limited FN emission regime and temperature-dependent regime. (c) FN plot of the measured emission current, showing non-constant slopes in the graphene emitter, attributed to its gate-sensitive band structure that allows dynamic tuning of the work function ϕ_{gr} and field enhancement factor β . (d) Schematic illustration of the charge-transport mechanism and current-saturation mechanism at the graphene edge, highlighting field emission, barrier modulation, and image-charge screening. (e) Emission current enhancement with increasing V_{GS} under a fixed V_{DS} , illustrating electrostatic modulation of the emission characteristics. The onset of current saturation, defined by the drain bias V_{sat} , exhibits a gate-dependent shift, indicating electrostatic control over the saturation behavior. All electrical measurements in this figure were obtained from a single fabricated device ($n = 1$). No statistical testing or p values apply, as these data represent deterministic physical device characteristics rather than population-based sampling.

distinct from solid-state SCLC conduction. In the SCL regime, the current increases with a power-law dependence on drain voltage (V_{DS}), while in the TD regime, the current gradually saturates with increasing voltage. Notably, the transition to the TD regime occurs at substantially lower drain bias ($\sim$ less than 40 V), in contrast to traditional vacuum diodes, where such behavior typically appears at several kilovolts [25]. This reduction in operating voltage is attributed to the sub-micron vacuum channel length achieved in our device.

In addition to the low operating voltage, the device exhibits enhanced current density, reaching approximately 0.3 mA/mm for a device width of 1 mm—more than 25 times higher than previously reported graphene edge emitters [26]. While this remains below the levels typical of solid-state devices such as GaN-based high-electron mobility transistors (HEMTs) [27, 28], it represents a significant step toward bridging the performance gap.

Further insight into the transport mechanism is obtained by analyzing the log-log slope of the I - V characteristics, defined

as $d\log(I_{DS})/d\log(V_{DS})$, as shown in Figure 2b. At lower drain voltages, the slope indicates field emission dominated by FN tunneling under space-charge-limited conditions. While classical SCL transport obeys the Child-Langmuir law ($I \propto V^{3/2}$) [24, 25], the observed emission exhibits a steeper, exponential dependence, indicative of FN-type behavior ($I \propto V^2/\phi_{gr} \exp(-\phi_{gr}^{3/2}/V)$) [29–31]. This deviation arises in part due to the semimetallic properties of graphene. Unlike ideal metals with a flat Fermi level and uniform field contours, graphene allows modulation of ϵ_F , which leads to non-uniform field distributions. According to our simulation study [1], under increasing drain bias, ϵ_F becomes spatially non-uniform near the emission edge due to electrostatic influence from the drain. This results in a field enhancement factor (β) that varies along the edge, as it becomes dependent on the local Fermi level distribution.

Additionally, electrostatic screening by accumulated carriers at the emission site reduces the penetration depth of the DWFCG field, diminishing its influence at high drain bias. However, the deviation in ϵ_F is accompanied by an increase in the local density of states (DOS), which in turn enhances the tunneling carrier

population. These competing effects—field screening versus increased tunneling availability—approximately balance each other out, thereby allowing the DWFCG to maintain effective modulation of the emission current over the entire tested voltage range.

Further insights into the field emission behavior of the device are provided by the FN plot, shown in Figure 2c. The FN representation, a plot of $\ln(I_{\text{DS}}/V_{\text{DS}}^2)$ versus $1/V_{\text{DS}}$, is commonly used to evaluate field emission according to the FN tunneling model, in which the emission current density J depends exponentially on the local electric field E , β , and the work function ϕ of the emitter [29, 30]. For an ideal metallic emitter with uniform geometry and constant ϕ , this plot yields a straight line.

However, in the present device, the FN plot deviates from linearity even at relatively low bias. This deviation reflects the bias-dependent and spatially varying nature of both β and ϕ_{gr} in the graphene emitter, consistent with the earlier discussion. As the drain voltage increases, space-charge effects induce electrostatic screening, which stabilizes both ϕ_{gr} and the local electric field distribution. Consequently, the emission current density saturates, and the FN plot begins to approximate linear behavior at higher bias.

Subtle dips in the log–log slope plot of Figure 2b provide additional evidence of the unique electronic structure of graphene. These features are attributable to non-monotonicities in the density of states (DOS) of graphene near the Dirac point, which influence the availability of tunneling electrons. Notably, the positions of these minima shift toward lower V_{DS} values as the gate bias increases. This trend is consistent with gate-induced modulation of the ε_F , which reduces the effective work function ϕ_{gr} at the emission edge and thereby lowers the tunneling barrier height. As a result of this modulation, the threshold voltage (V_{th}) for strong emission systematically decreases with increasing gate voltage. This behavior further confirms the role of electrostatic tuning of ε_F in controlling emission characteristics, and it illustrates the effectiveness of the DWFCG in enabling dynamic and efficient gate control in graphene-based NVTs.

Within the TD regime, two distinctive and technically significant features are observed. First, the emission current exhibits a remarkably flat saturation behavior, as evident in Figure 2a,b, while remaining effectively modulated by the DWFCG bias. In conventional vacuum diodes, the TD regime typically shows a gradual increase in current, often attributed to Schottky barrier lowering induced by image charge effects [24]. As more charge accumulates at higher drain voltages, the resulting image potential lowers the emission barrier, thereby enhancing the current.

In contrast, as shown in Figure 2d, graphene edge emitters experience inherently weak image-charge formation because the atomically thin edge cannot sustain a well-defined mirror plane for classical screening [32, 33]. As a result, the induced image potential becomes spatially diffuse rather than localized, substantially reducing the extent of Schottky barrier lowering under high drain fields [32–36]. Combined with graphene's finite quantum capacitance—which further limits

localized charge accumulation—this weakened screening keeps the emission barrier comparatively stable. This stabilized barrier, together with the gradual field attenuation caused by space-charge buildup in the vacuum channel, produces the extended and smooth saturation behavior observed in our device.

Remarkably, the saturation current level is also tunable by the DWFCG alone, as shown in Figure 2a. When the ε_F of the graphene edge is raised via DWFCG biasing, the unique linear energy–momentum dispersion of graphene implies a shift toward higher in-plane momentum ($\hbar k$) states. This shift enhances the density of available tunneling carriers, contributing to the increased saturation current. Functionally, this resembles tuning the emitter material's work function—an effect that, while unintuitive in conventional systems, becomes feasible due to graphene's dualistic nature, bridging semiconducting and metallic behavior.

Further evidence of DWFCG control is observed in Figure 2e, which shows that the drain voltage at which current saturation occurs (V_{sat}) decreases systematically with increasing gate bias. This shift results from the electrostatic modulation of both ε_F and ϕ_{gr} at the graphene edge. A more positive gate voltage raises ε_F , lowers the effective work function ϕ_{gr} and enhances the tunneling probability. This accelerates space-charge accumulation at lower drain voltages, thereby inducing earlier onset of saturation. The observed dependence of V_{sat} on DWFCG bias illustrates the interaction between the drain-induced longitudinal field and the DWFCG-induced transverse field at the emitter.

While the dominant saturation behavior follows field emission theory and gate-modulated ϕ_{gr} , small oscillations appear in the emission current under steady-state bias. These fluctuations likely stem from thermal perturbations in the space-charge cloud [37–42]. In the quasi-ballistic regime, where scattering is minimal, electron–electron interactions drive charge redistribution, and thermal fluctuations in carrier density and local fields can modulate the emission barrier dynamically. Though the exact origin is uncertain, these oscillations suggest many-body effects in ballistic vacuum transport and warrant further study.

Although simulated and measured current densities align in magnitude, the simulations do not fully capture the observed saturation behavior. This stems from simplified assumptions, such as neglecting space-charge effects and adopting idealized geometry with thinner dielectrics and flat emitter surfaces [1], which limit accuracy under high-field conditions. Introducing a TD regime beyond a critical field can enhance the model's ability to reflect saturation phenomena.

3.2 | Operation at Extremely Low Temperatures

Vacuum electronic devices maintain reliable operation under extreme temperatures, unlike solid-state transistors, which fail at cryogenic conditions due to dopant freeze-out. Since they do not rely on thermally generated carriers from impurity atoms within a solid-state channel, vacuum transistors can remain functional at low temperatures. This robustness enables detailed investigation

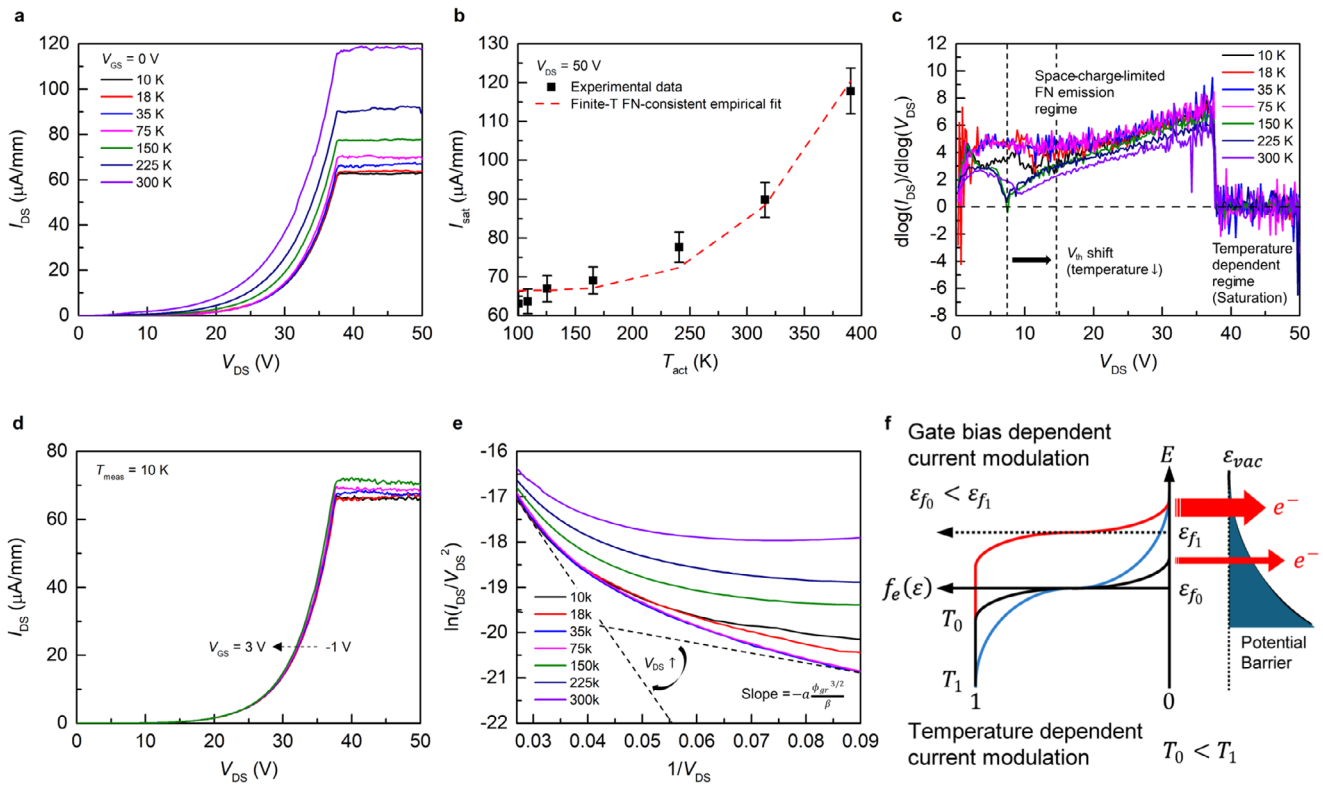


FIGURE 3 | (a) Emission current as a function of V_{DS} measured at various temperatures (10 to 300 K), showing a strong temperature dependence driven by thermal broadening of the Fermi–Dirac distribution. (b) Saturation current plotted against self-heating-corrected temperature T_{act} , fitted using a hybrid FN–RD model combining tunneling and thermionic emission. (The error bars represent $\pm 5\%$.) (c) Logarithmic derivative $d\log(I_{DS})/d\log(V_{DS})$ plotted against V_{DS} at various temperatures. The emission current is highly temperature-dependent; with increasing temperature, the emission current rises due to thermal excitation, while enhanced space-charge accumulation in the vacuum channel strengthens electrostatic screening and reduces the current’s sensitivity to V_{DS} . (d) I – V characteristics of the graphene-based NVT measured at 10 K. (e) FN plot of the measured emission current at various temperatures. The temperature-dependent variation in instantaneous slope arises from the graphene band structure at each temperature and the screening effect caused by space charges. At lower temperatures, the electric field induced by the drain bias exerts a more significant influence under high V_{DS} conditions. (f) Schematic illustration of two distinct mechanisms by which temperature variation and gate modulation influence the band structure. Both mechanisms independently contribute to an increase in high-momentum electron population, resulting in modulation of the saturation current. All electrical measurements in this figure were obtained from a single fabricated device ($n = 1$). Error bars represent measurement uncertainty ($\pm 5\%$) rather than statistical variance. No statistical testing or p values apply, as these data reflect deterministic physical device behavior rather than population-based sampling.

of emission behavior across a wide temperature range, revealing key aspects of electron transport and field emission mechanisms.

The measured I – V characteristics from $T_{meas} = 10$ to 300 K, shown in Figure 3a, reveal a systematic reduction in emission current at lower temperatures, consistent with expectations. At higher temperatures, thermal broadening of the Fermi–Dirac distribution increases the number of high-energy electrons, which face thinner tunneling barriers, thereby enhancing assisted FN tunneling thermally.

In the TD regime, the emission current partially follows the Richardson–Dushman (RD) equation, which models thermionic emission [24, 25]. As shown in Figure 3b, the experimental data fit this model well, particularly at elevated temperatures. A more accurate description is provided by a hybrid expression, $I(T_{act}) = I_{FN} + AT_{act}^2 \exp[-\phi_{gr}/(k_B T_{act})]$, where $T_{act} = T_{meas} + \Delta T$. Under high-field conditions, this model reflects the coexistence of a finite-temperature Fowler–Nordheim (finite-T FN) component—which remains weakly temperature dependent—

and a field-lowered RD contribution associated with thermally assisted emission [29, 31]. Global fitting yields $I_{FN} = 6.65 \times 10^{-5}$ A, $\Delta T \approx 90.6$ K, and $R^2 = 0.964$, indicating superior agreement across the wide temperature range.

Under high-temperature conditions, the balance between field-enhanced tunneling and thermally assisted emission is expected to shift toward the latter, which naturally weakens the effectiveness of gate-induced Fermi-level modulation in graphene. Nevertheless, the vacuum environment and the intrinsic thermal stability of graphene edges suggest that the device can remain operational well above room temperature. These considerations indicate that while gate control may gradually diminish at elevated temperatures, the overall emission process should remain robust.

The elevated emitter temperature, T_{act} , reflects local self-heating from Joule dissipation, field emission losses, and carrier relaxation, and is consistent with the large thermal boundary resistance of the van der Waals (vdW) interface. Deviations from

ideal RD behavior arise from graphene's weak image-charge formation and bias-dependent barrier profile, which modify the effective emission barrier at finite fields [32, 33]. Importantly, the apparent quadratic temperature dependence ($\propto T^2$) of the saturation current at low temperatures does not imply classical thermionic emission. At cryogenic temperatures, the Richardson–Dushman mechanism is negligible, and the observed T^2 scaling instead reflects finite-temperature (thermally assisted) FN tunneling. This behavior is fully consistent with the Murphy–Good finite-temperature correction, where non-zero temperature broadens the Fermi–Dirac supply function without invoking true thermionic emission. To prevent misunderstanding, we interpret this regime as finite-T FN tunneling rather than a hybrid FN–thermionic process.

Closer examination of the low-temperature data (Figure 3c) reveals two subtle but notable features: (i) the disappearance of characteristic dips—local minima observed in the log–log slope—at temperatures below approximately 75 K, (ii) a shift in emission onset toward higher drain biases as temperature decreases. These effects are attributed to a reduced population of thermally excited electrons and a narrower energy distribution near the ε_F . As thermal excitation diminishes, fewer electrons have sufficient energy to tunnel through the potential barrier. The observed increase in the log–log slope of the I – V curve at low temperatures is thus due to a reduction in baseline current, rather than enhanced emission efficiency.

Beyond intrinsic low-temperature behavior, changes in I – V curves before (Figure 2a) and after (Figure 3a) cryogenic cycling indicate a reduced emission current under identical bias. This suggests thermomechanical stress may have caused structural or alignment shifts, affecting local field enhancement and gate–emitter coupling. Such thermal strain likely originates from a combination of global substrate contraction during cooling and localized Joule heating at the graphene edge during field emission. As previously estimated from the RD fitting, the emitter temperature can be ~ 90.6 K higher than the substrate due to self-heating. This thermal gradient across nanoscale features can induce mechanical stress, particularly at the gate–emitter interface. Although gradual cryogenic cycling (300→10 K over six hours; return over two), some thermal stress still seemed unavoidable.

Despite these effects, the device continued to exhibit emission behavior consistent with FN tunneling at all tested temperatures, with current modulation preserved via the DWFCG. As shown in Figure 3d, the device operated reliably at 10 K, albeit with reduced modulation depth compared to room temperature. The corresponding FN plot in Figure 3e confirms that both β and ϕ_{gr} vary systematically with temperature, even though the device geometry and applied bias were held constant.

The temperature dependence reveals two key emission mechanisms: (i) thermal redistribution of carriers: at elevated temperatures, a broader occupation of high-energy electron states modifies the tunneling distribution. (ii) space-charge effects: changes in electron density within the vacuum channel alter the electrostatic screening environment, which influences the effective field at the emitter surface. These two emission control mechanisms are summarized in Figure 3f: (i) thermal distri-

bution, which influences the overall current level, and (ii) DWFCG-induced modulation, which dominates both SCL and TD regimes.

The proposed architecture enables effective saturated current control via electrostatic band tuning at the graphene edge—a function traditionally requiring multielectrode vacuum tubes like tetrodes [25]. Using a single off-channel gate, the design achieves current saturation with high output impedance. The high output impedance can aid signal isolation and impedance matching in RF circuits. The resulting I – V characteristics indicate strong potential for high-frequency amplifier use, particularly in cryogenic or space environments. By avoiding complex multi-gate integration, this architecture delivers tetrode-level performance in a compact, scalable form.

3.3 | Circuit-Level Feasibility Analysis Based on Measured Device Parameters

To assess whether the proposed work-function modulation scheme enables functional amplification behavior beyond conceptual demonstration, amplifier circuits were numerically evaluated using a SPICE model based on measured output characteristics (Figure 2a, $V_{GS} = -1$ to 3 V). As SPICE lacks self-consistent electrostatics, output currents were fitted via polynomial regression, enabling direct extraction of small-signal parameters such as transconductance (g_m) and output resistance (r_o). Circuit details are shown in Figure 4a; modeling procedures are in [Supporting Information](#). A full circuit-level demonstration requires a vacuum-compatible test platform with controlled parasitics, feedthrough structures, and stabilized packaging, which lies beyond the scope of this first device-level study; the present SPICE analysis is therefore intended as a feasibility assessment based directly on measured device characteristics rather than a substitute for an experimental amplifier.

For clarity, the amplifier results presented here are numerical projections based exclusively on experimentally extracted small-signal parameters, included to illustrate the intrinsic feasibility enabled by the demonstrated work-function modulation. They are not intended as a realized or optimized circuit implementation.

As shown in Figure 4b, load-line analysis highlights a clear contrast between the SCL and TD regimes. In the SCL regime (load resistance, $R_L = 400$ k Ω), the device shows $r_o \approx 35.7$ k Ω and a voltage gain (A_{v0}) of 4.2 V/V. Under saturation ($R_L = 200$ k Ω), r_o increases to ~ 3.11 M Ω , yielding a much higher gain of $A_{v0} = 50$ V/V. While functional amplifier design is feasible even within the SCL regime, the emergence of a well-defined current saturation region introduces the potential for substantially enhanced gain. This makes the saturation regime particularly attractive for achieving higher intrinsic gain in circuit-level projections, while a full assessment of bias dependence and stability would require dedicated circuit-level test structures beyond the scope of this work.

Figure 4c presents the circuit diagram of a single-stage common-source (CS) amplifier incorporating the graphene NVT. This amplifier yields a projected voltage gain of 10.4 dB (3.31 V/V)

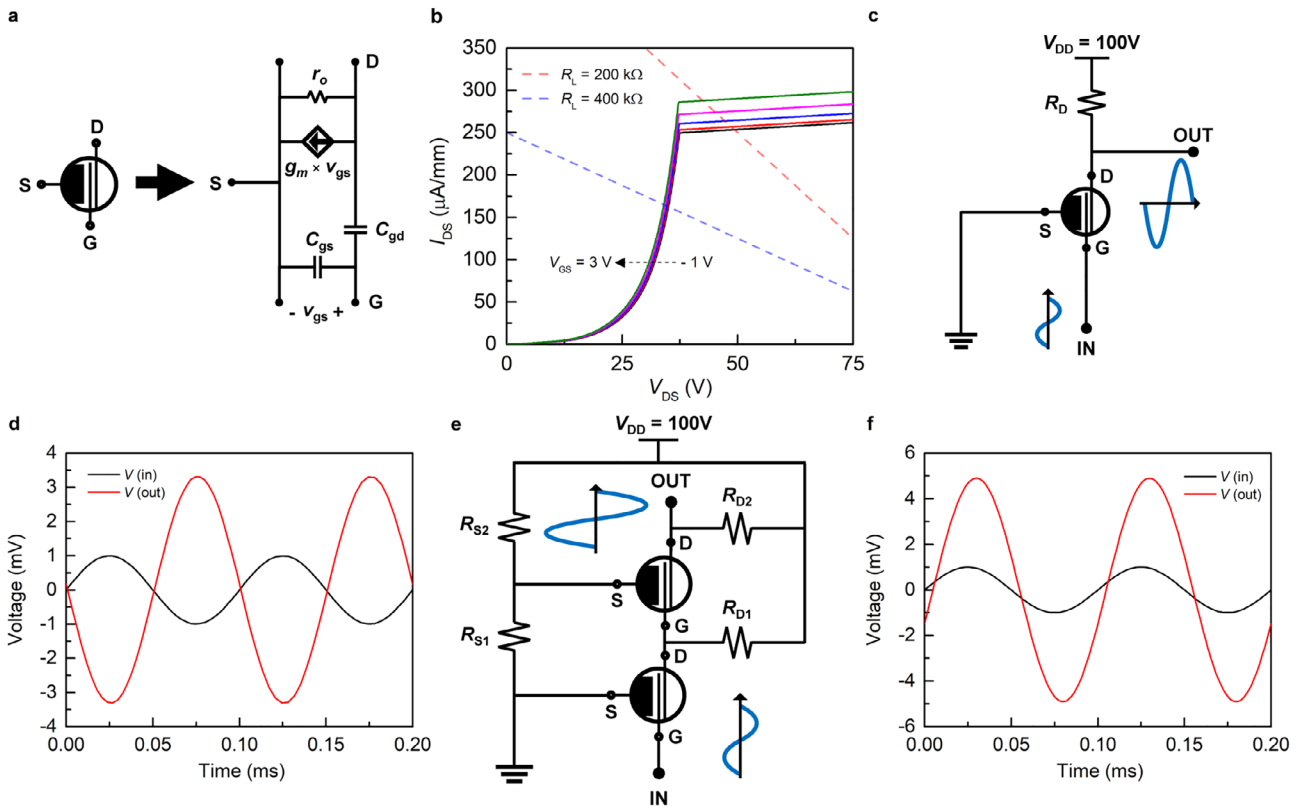


FIGURE 4 | (a) SPICE symbol of the proposed NVT and equivalent hybrid- π model for small-signal analysis. (b) Modeled output characteristics of the vacuum channel transistor with load-line analysis for 400 and 200 k Ω . At 400 k Ω , the device operates in the exponential regime with low output resistance and limited gain (4.17 V/V). At 200 k Ω , it operates in the saturation regime with high output resistance and enhanced gain (50.3 V/V). (c) Schematic of the single-stage amplifier circuit designed for gain evaluation, with $R_D = 220 \text{ k}\Omega$ for stable saturation operation. (d) Input and output waveforms of the single-stage amplifier. (e) Schematic of the two-stage amplifier circuit with bias stabilization using a resistive voltage divider. $R_{D1} = 220 \text{ k}\Omega$, $R_{D2} = 99 \text{ k}\Omega$, $R_{S1} = 10 \text{ k}\Omega$, and $R_{S2} = 19.4 \text{ k}\Omega$ were used to maintain saturation in both stages. (f) Input and output waveforms of the two-stage amplifier. All data in this figure were obtained from a single SPICE simulation model ($n = 1$). No statistical testing or p values apply, as these results reflect deterministic circuit simulations rather than population-based sampling.

under a supply voltage $V_{DD} = 100 \text{ V}$. The output waveform, obtained from the numerical simulation with a sinusoidal input signal of $V_{GS} = 2.95 \pm 1 \text{ mV}$ at 10 kHz, is shown in Figure 4d.

In a two-stage amplifier configuration (Figure 4e), the overall gain is projected to increase to 13.8 dB (4.9 V/V), consistent with Class-A behavior in the simulated response (Figure 4f). The first-stage output directly drives the second-stage gate, with a gate bias fixed at 2.95 V. A resistive voltage divider ensures that the second-stage gate voltage remains within the operational range (−1 to 3 V). Because the external load resistors R_L are relatively smaller than the device's intrinsic output resistance, the observed gain is lower than the intrinsic A_{v0} . However, increasing both V_{DD} and R_L significantly improves amplifier gain in the parameter-based simulation. As shown in Table 1, gains of 33.7 V/V at $V_{DD} = 200 \text{ V}$ (with $R_L = 570$ and 445 k Ω), and up to 134 V/V at 500 V (with $R_L = 1.6$ and 1.4 M Ω) were obtained in simulation. These results illustrate the sensitivity of the projected gain to biasing and loading conditions in multi-stage configurations.

The relatively low g_m is mainly attributed to the thick gate dielectric and extended emitter edge, both chosen to minimize average gate leakage ($< 10^{-12} \text{ A}$). Thinner high- k dielectrics (2–3 nm) could significantly enhance gate coupling. Shortening the

vacuum channel and refining emitter geometry would further boost field concentration and reduce space-charge effects. While performance optimization was not the focus here, the device already shows high output resistance—critical for amplifiers—and future structural improvements may enable higher gain and better linearity.

Table 2 compares recent nanoscale vacuum transistors, highlighting the trade-off between high on/off ratio and ultra-low gate leakage—rarely achieved together due to structural constraints.

Gate-all-around designs (e.g., Ref. [15]) offer strong electrostatic control but are difficult to scale, while bottom-gate architectures (e.g., Ref. [46]) simplify fabrication at the cost of weaker coupling and limited RF suitability. Prior graphene-based plane or “in-channel” designs (e.g., Ref. [43]) primarily reshape the vacuum-gap field and can experience gate–beam interference and limited current saturation, and point/tip emitters (e.g., Ref. [15] and Ref. [45]) achieve high local fields but do not exploit graphene's electrostatically tunable Fermi level for work-function–based modulation. Even planar graphene nanogap configurations (e.g., Ref. [46] and Ref. [47]) and back-gated graphene edge emitters (e.g., Ref. [5]) modulate only the channel field rather than the emitter's electronic state, resulting in geometric field enhancement

TABLE 1 | Voltage gain analysis results according to the supply voltage (V_{DD}) and load resistances (R_{D1} , R_{D2}). each parameter corresponds to the values set in the schematics of the single-stage cs amplifier in Figure 4c and the two-stage cs amplifier in Figure 4e. all values represent numerical projections derived from the parameter-based spice analysis. It can be observed that as V_{DD} increases, larger load resistance values can be applied, resulting in improved voltage gain of the circuit.

V_{DD}	The number of Stage	R_{S1}	R_{S2}	R_{D1}	R_{D2}	A_v (Voltage gain)
100 V	Single	—	—	220 k Ω	—	3.31 V/V (10.4 dB)
	Two	10 k Ω	19.4 k Ω	220 k Ω	99 k Ω	4.9 V/V (13.8 dB)
200 V	Single	—	—	570 k Ω	—	7.76 V/V (17.8 dB)
	Two	10 k Ω	48.7 k Ω	570 k Ω	445 k Ω	33.7 V/V (30.5 dB)
500 V	Single	—	—	1.6 M Ω	—	16.9 V/V (24.5 dB)
	Two	10 k Ω	121 k Ω	1.6 M Ω	1.4 M Ω	134 V/V (36.5 dB)

TABLE 2 | Comparison of key electrical characteristics for various vacuum emitter configurations, including planar, point, and line-type emitters. The table summarizes the V_{GS} , V_{DS} , on current (I_{on}), on/off ratio (I_{on}/I_{off}), and on current to gate leakage current ratio (I_{on}/I_G) for each device. The graphene edge emitter device developed in this work, featuring a top-gate structure, demonstrates competitive on current and exceptionally high I_{on}/I_G ratio, highlighting its advantages for stable, gate-controlled field emission in nanoscale vacuum transistors.

Emitter dimension	Configuration	On V_{GS} [V]	Off V_{GS} [V]	V_{DS} [V]	I_{on} [μ A]	I_{on}/I_{off}	I_{on}/I_G	Gain [V/V]	Refs.
Plane	Graphene plane emitter n+ Si collector	5.6	4	15	7.2×10^3	10^6	1.04×10^{-5}	—	[43]
Point	Gate-all-around Si nanowire tip	2	0	5	3	4×10^3	$> 10^6$	—	[15]
	Gate under metal tip emitter/collector	60	− 60	35	32	1.65	$> 10^4$	—	[44]
	Si wafer gate under tip vertex emitter/collector	0.9	0	1.7	0.2	141	—	—	[45]
	Si wafer gate under metal emitter & 3-way collectors	6	− 6	7	17	10^4	944	—	[46]
	Metal tip side-gate near metal tip emitter/collector	5.8	0	15	0.35	10^3	10	—	[47]
Line	Si wafer gate under graphene nano gap	18	4	7.5	0.25	100	3×10^5	—	[5]
	Top-gate on graphene edge emitter	3	− 1	$6.3 \sim 4.7$ (V_{th})	$291 (I_{sat})$	$2.24 \geq$	490×10^3	$4.9 (V_{DD} = 100 \text{ V})$	This work

instead of genuine work-function control. In contrast, our device employs a laterally aligned off-channel gate that couples directly to the graphene edge and shifts its Fermi level, providing an intrinsic and scalable mode of emission modulation without intersecting the electron trajectory.

Building on this distinction, our top-gate, top-contact configuration places the control electrode in direct lateral alignment with the graphene edge while remaining outside the electron path. This off-channel geometry delivers strong electrostatic coupling with reduced parasitic capacitance and eliminates the gate–beam interception issues encountered in in-channel or back-gated structures. The resulting vertical gate field enables efficient work-

function modulation without added fabrication complexity and is compatible with scaling toward high-frequency operation. Collectively, these features provide a balanced combination of gate efficiency, leakage suppression, and scalability, addressing the structural and mechanistic limitations observed in prior nanoscale vacuum transistor designs.

4 | Conclusion

We have realized a graphene-based nanoscale vacuum transistor in which an off-channel DWFCG enables robust, low-leakage current modulation and clear current saturation over a broad temperature range, from 300 down to 10 K. This capability, arising

from electrostatic Fermi-level tuning unique to graphene, allows dynamic control of emission onset and saturation without intercepting the electron beam—achieving key functional features analogous to tetrode-like architectures in a compact and scalable device geometry. By decoupling emission and control fields, this approach addresses fundamental limitations of conventional metallic emitters and in-channel-gated nanoscale vacuum transistors, and establishes direct work-function modulation as a viable mechanism for vacuum electronic devices. While further optimization and system-level development are required, the demonstrated platform provides a foundation for exploring vacuum electronic operation in regimes where precise emission control, low leakage, and environmental robustness are of interest, including cryogenic and radiation-tolerant applications.

5 | Experimental Section/Methods

5.1 | Device Fabrication Process

The graphene-based NVT was fabricated using commercially available monolayer graphene (Graphenea), transferred onto a 150 nm-thick Si_3N_4 dielectric layer deposited via Low-Pressure Chemical Vapor Deposition (LPCVD) on a (100)-oriented silicon substrate. The sheet resistance of graphene, measured using a four-point probe system (CMT-SR2000N, Advanced Instrument Technology), was approximately $550 \, \Omega/\square$.

An Al_2O_3 gate dielectric layer with a thickness of 40 nm was deposited over the graphene sheet using Atomic Layer Deposition (ALD; Lucida D200, NCD) at 200°C . To ensure optimal oxide quality and minimize thermal stress, the sample was first subjected to a controlled temperature ramp-up at a rate of $1^\circ\text{C}/\text{min}$ until reaching 300°C , then annealed for 30 min under a nitrogen (N_2) atmosphere (100 sccm, 1 Torr), and subsequently cooled naturally to 50°C .

Subsequent device patterning involved standard photolithography techniques using a micro-pattern generator (μMLA , Heidelberg Instruments). Initially, alignment keys were patterned to ensure precision in subsequent steps. The mesa region was defined by selectively etching unwanted areas using both wet and dry etching methods. Buffered oxide etchant (BOE 1:10) was employed for wet etching, while Inductively Coupled Plasma-Reactive Ion Etching (ICP-RIE; KVET-I200L, Korea Vacuum Tech) was utilized for dry etching with Ar and CF_4 plasmas for oxide removal and Ar and O_2 plasmas for graphene etching.

Photoresist (PR) processing utilized LoR5A (MicroChem) as the lift-off resist and S1805 (MicroChem) as the positive resist, both spin-coated at 3000 RPM for 30 s. LoR5A was soft-baked at 180°C for 250 s, and S1805 was baked at 115°C for 2 min, each followed by ambient cooling. Adhesion enhancement was achieved using Surpass 4000 (MicroChem) prior to spin-coating ma-N 1440 (Microresist Technology), which was soft-baked at 100°C for 4 min.

Following lithography, development employed TMAH 2.38% and ma-D 533D/s (Microresist Technology), and PR removal was performed with mr-rem 700 (Microresist Technology). Metal electrodes consisting of Pd (40 nm)/Ti (10 nm) were deposited

using electron beam evaporation (20KVS037, Korea Vacuum Tech). The vacuum channel was precisely defined via a secondary lithographic step with intentional PR over-development, followed by dry etching. Finally, graphene emitter edges were exposed and precisely protruded by controlled hydrofluoric acid vapor (Idonus) etching, which selectively removed the insulating layer in the direction opposite to emission, thereby revealing the edges and optimizing electron emission.

5.2 | Electrical Measurements

Electrical measurements were conducted using a Keithley 4200-SCS parameter analyzer configured in a common-source arrangement. Precise thermal control, particularly under cryogenic conditions, was achieved using a CRX-VF cryogenic probe station (Lake Shore). High vacuum ($\sim 10^{-5}$ mbar) conditions were maintained throughout testing using a TPS-compact Turbo Pumping System (Agilent). The device's electrical characteristics were assessed using an independently contacted three-terminal configuration (source, gate, and drain).

5.3 | Device Characterization

Structural and compositional characterization employed advanced microscopy techniques. A protective Pt layer was deposited to enhance surface conductivity and protect device structures during subsequent processing. Regions of interest for cross-sectional Transmission Electron Microscopy (TEM) were precisely defined and protected using a localized gas-assisted electron beam deposition process, followed by an additional protective ion beam-deposited layer.

Sample thinning and milling were performed using a Helios 600 focused ion beam (FIB) system (FEI, Bruker), carefully thinning the sample to electron transparency while preserving structural integrity. High-resolution imaging and electron energy loss spectroscopy (EELS) mapping were conducted using a spherical aberration-corrected JEM-2200FS TEM (JEOL) at 200 kV, enabling detailed analysis of graphene quality and dielectric interfaces.

Raman spectroscopy, performed using an Alpha300 R system (Oxford Instruments), assessed graphene integrity and confirmed minimal structural degradation pre- and post-ALD process, thus validating the overall fabrication quality and consistency.

5.4 | Statistical Analysis

Electrical measurements were performed on devices ($n = 2$), and the presented characteristics correspond to representative behavior observed across the measured samples. No additional statistical hypothesis testing was required, as the study focuses on device-level electrical transport rather than population-based comparisons. Raw current-voltage data were inspected to confirm the absence of measurement artifacts, and no pre-processing other than baseline correction was applied. Data were plotted using OriginPro and MATLAB, and values are reported as directly measured.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available in the Supporting Information of this article.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File: sml172093-sup-0001-SuppMat.docx